Pick&Place con TIAGo e RoboticsToolbox

Fondamenti di Robotica

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Outline

- RoboticsToolbox – Python
- Esercitazione per task 2

Robotics Toolbox – Python

- Libreria sviluppata per facilitare la simulazione e il controllo di robot nell'ambiente Python
- È parte del Progetto "**Robotics Toolbox**" creato da Peter Corke

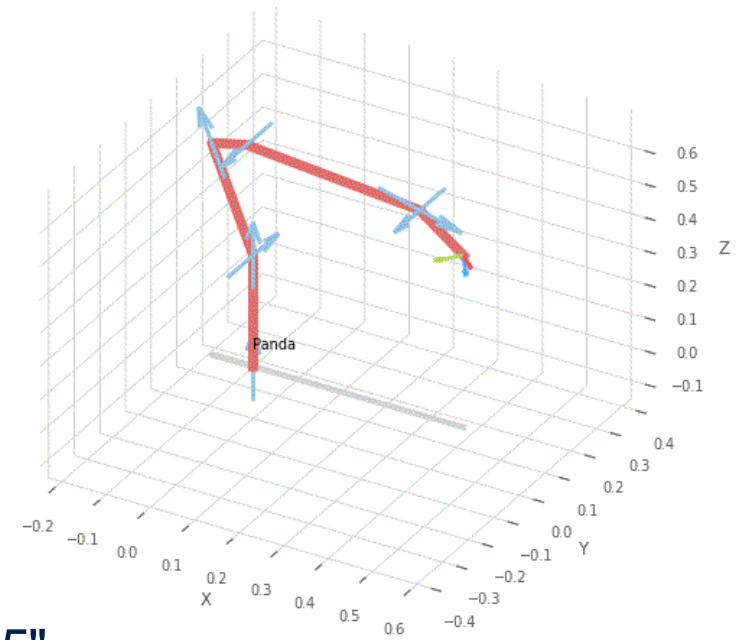
Documentazione:

- https://github.com/petercorke/robotics-toolbox-python/tree/master
- https://petercorke.github.io/robotics-toolbox-python/intro.html

Tutorial: https://github.com/jhavl/dkt

Caratteristiche

- Supporto per robot manipolatori
- Cinematica diretta e inversa
- Dinamica del robot
- Generazione di traiettorie
- Simulazione in 2D e 3D
- Supporto per ROS




```
python3 -m pip install --user --upgrade "numpy<1.25"
```

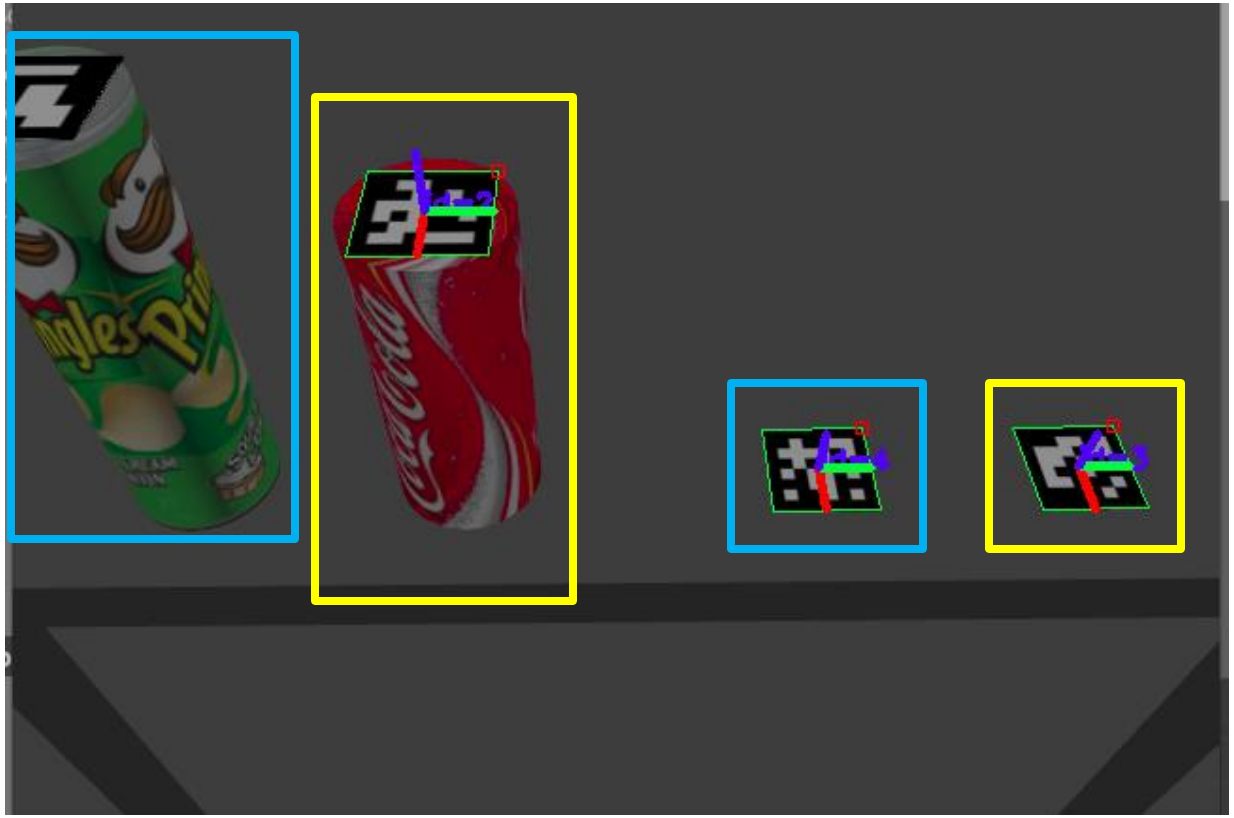
Esercitazione per task 2

Pick & place degli oggetti

→ publisher, subscriber, plugin Gazebo, service, action, rotazioni, RoboticsToolbox, Rviz, Gazebo, tf

→ Pick Coca Cola con gestione sequenza

- TO DO per il task 2:
  - ❑ Place Coca Cola
  - ❑ Pick Pringles
  - ❑ Place Pringles
  - ❑ Gestione sequenza
  - ❑ Launch



Esercitazione per task 2

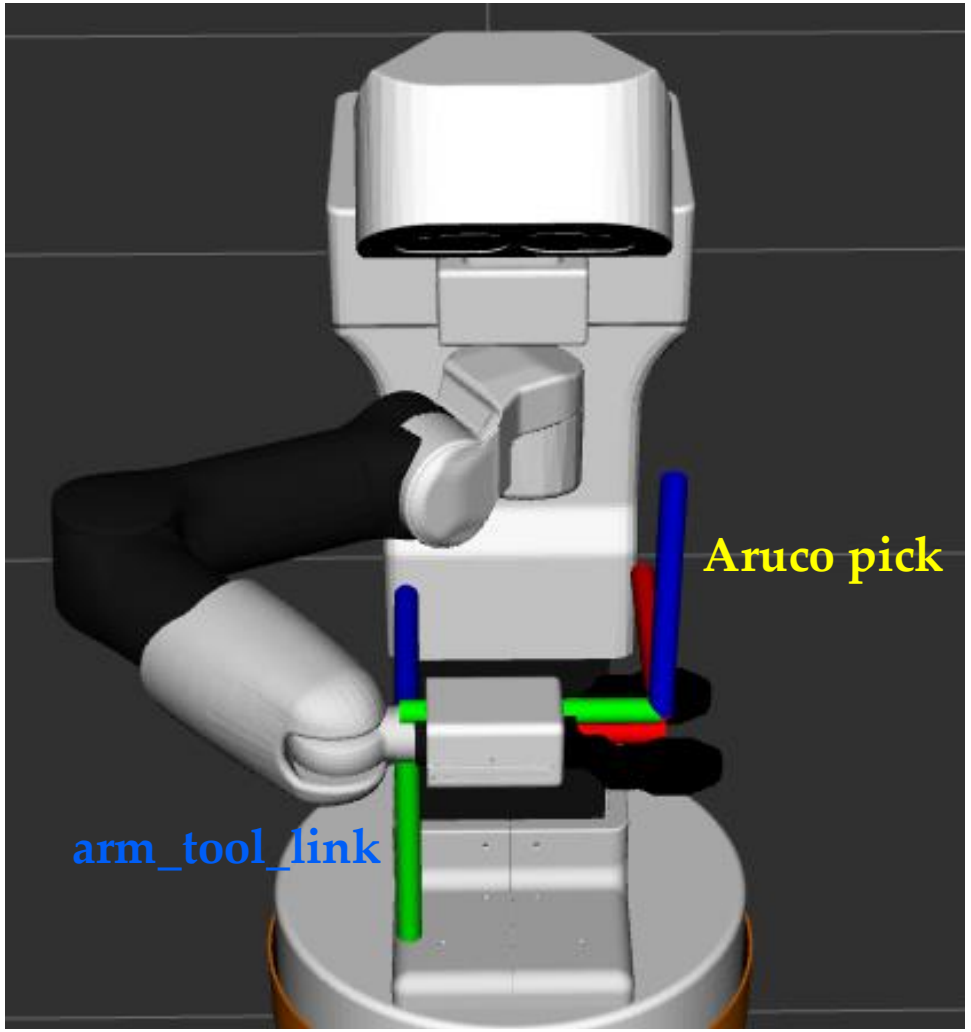
Caricamento del modello

- Scaricare il modello tiago_robot.urdf da elearning
- Inserire il file appena scaricato in un pkg «my_robot_description\urdf» all'interno del workspace del progetto
- Creare un nuovo package «pick_place»
- Nel pkg creato, creare un file pick_place_cocacola.py
- Caricare il modello urdf nel codice.py

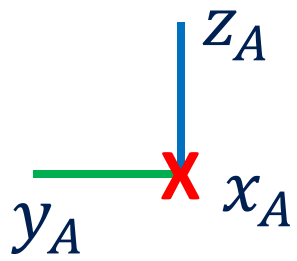
```
from roboticstoolbox import ERobot

urdf_loc =
'/home/rita/FdR_ISI2026/esercitazioni_2026/src/my_robot_description/urdf/tiago_robot.
urdf'
robot = ERobot.URDF(urdf_loc)
```

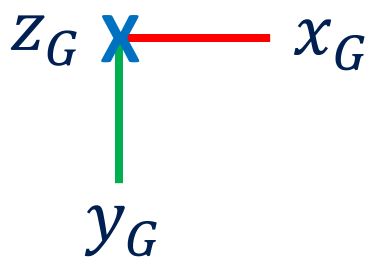
Pick Coca Cola



ARUCO PICK

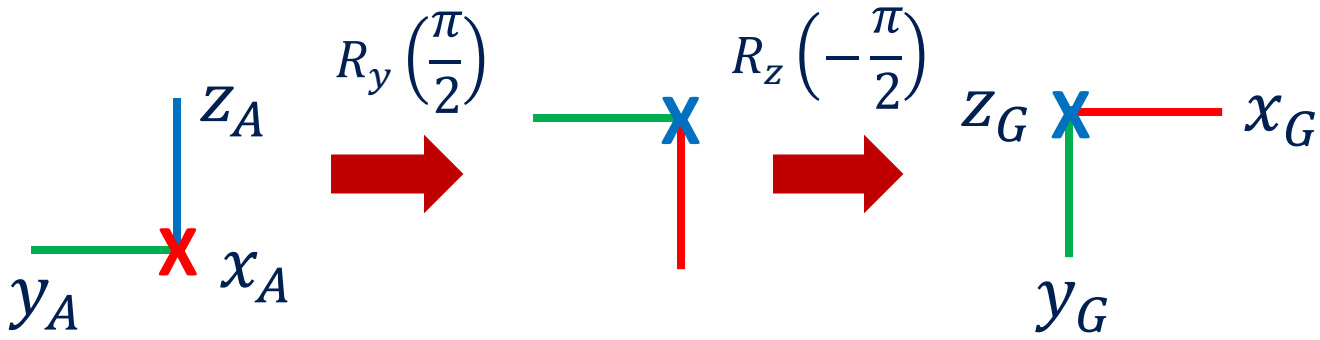


GRIPPER PICK



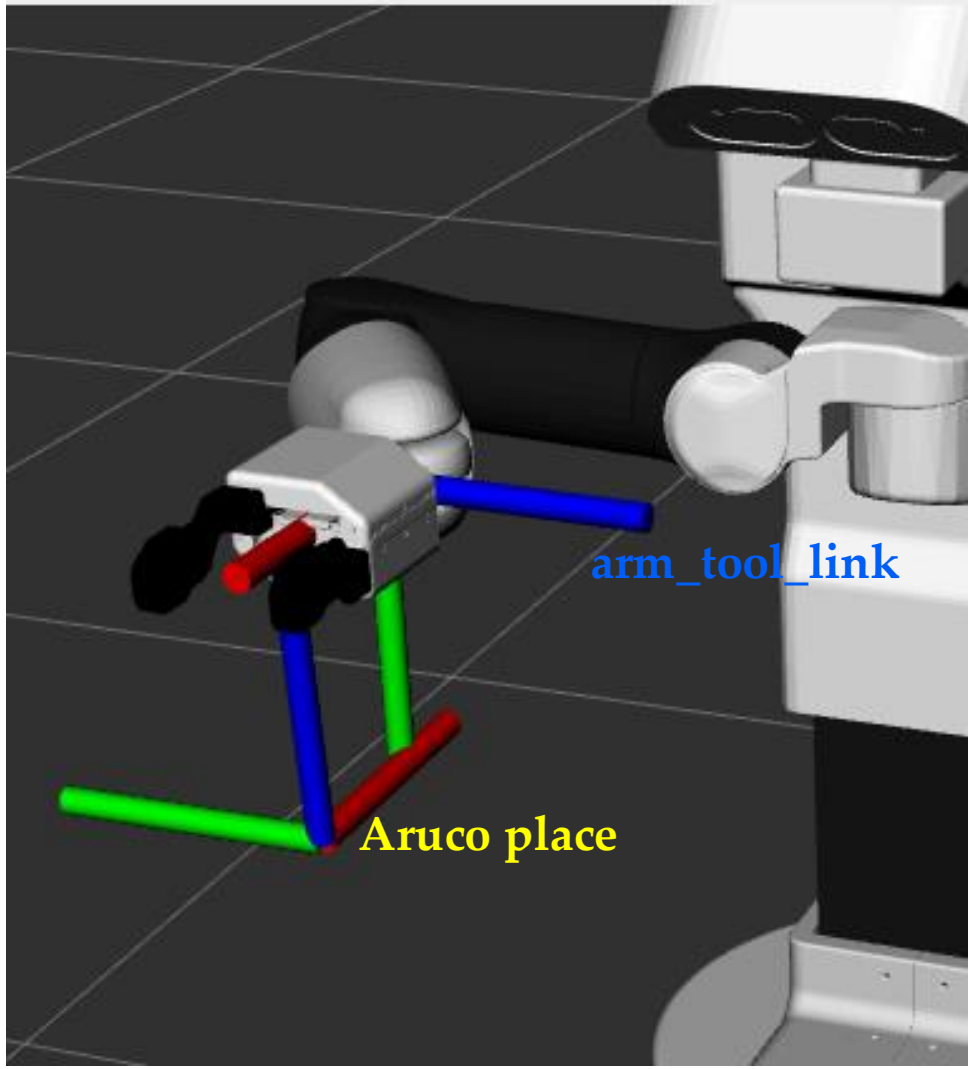
$$x_G = -y_A$$
$$y_G = -z_A$$
$$z_G = x_A$$

$$R_G^A = \begin{matrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{matrix}$$

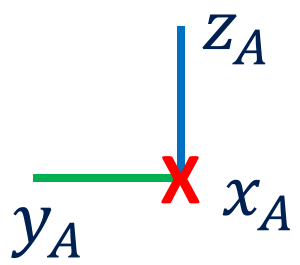


T_pick = SE3.Ry(np.pi/2)*SE3.Rz(-np.pi/2)

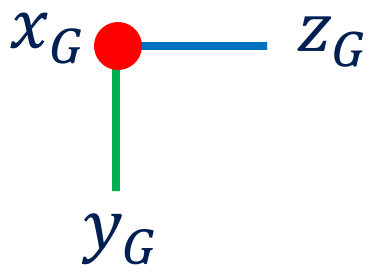
Place Coca Cola



ARUCO PLACE

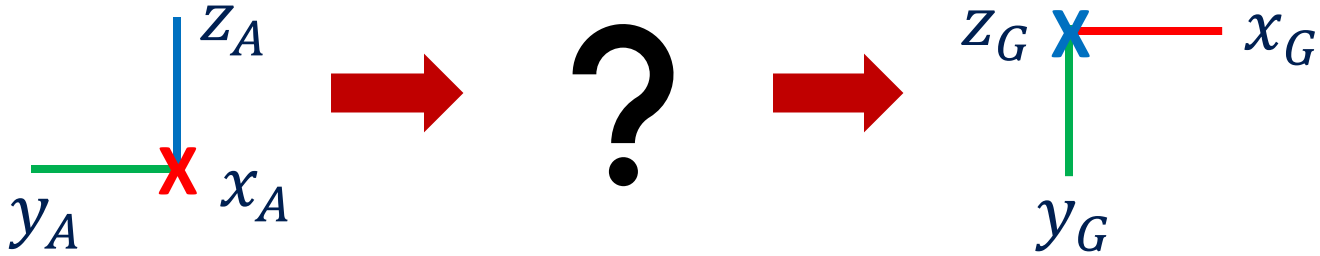


GRIPPER PLACE



$$x_G = -x_A$$
$$y_G = -z_A$$
$$z_G = -y_A$$

$$R_G^A = \begin{matrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{matrix}$$



T_place = ?????

Esercitazione per task 2

Terminali:

- ros2 launch tiago_gazebo tiago_gazebo.launch.py is_public_sim:=True
- ros2 run aruco_pkg aruco_detection
- ros2 run aruco_pkg action_client
- ros2 run aruco_pkg action_server
- ros2 run play_motion2 run_motion unfold_arm
- ros2 run pick_place_coca pick_place_cola